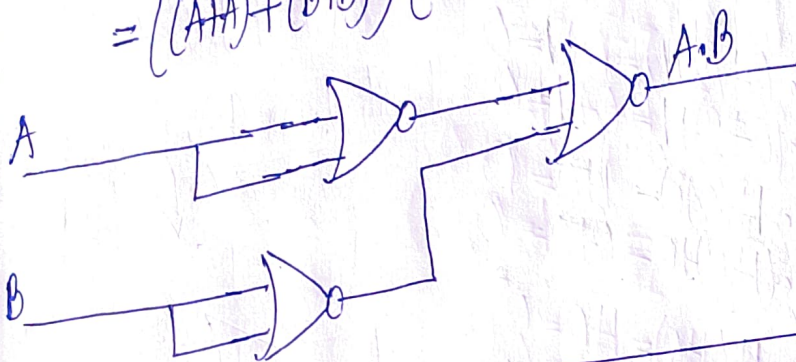


- The exclusive-OR is an odd function, i.e. it is equal to 1 if the input variables have an odd number of 1s.
- The equivalence function is an even function, i.e. it is equal to 1 if the input variables have an even number of 0s.

Example 2-7: Construct AND Gate using NOR gate

$$\begin{aligned}
 A \cdot B &= ((A \cdot B)')' \quad (\text{Theorem 3}) \\
 &= (A' + B')' \quad (\text{Theorem 5(b)}) \\
 &= ((A + A') + (B + B'))' \quad (\text{Theorem 1(a)})
 \end{aligned}$$



Problems

2-1: Which of the six basic laws (closure, associative, commutative, identity, inverse and distributive) are satisfied for the pair of binary operators listed below?

+	0	1	2
0	0	0	0
1	0	1	1
2	0	1	2

•	0	1	2
0	0	1	2
1	1	1	2
2	2	2	2

Solution: The given set is $\{0, 1, 2\}$. It's easy to see that for both + and •, the closure operator holds.

- $0+0=0$, $0+1=1+0=0$, $0+2=2+0=0$, $1+1=1$, $1+2=2+1=1$
- $2+2=2$, $\therefore +$ follows the commutative law
- $0 \cdot 0=0$, $0 \cdot 1=1 \cdot 0=1$, $0 \cdot 2=2 \cdot 0=2$, $1 \cdot 1=1$, $1 \cdot 2=2 \cdot 1=2$
- $2 \cdot 2=2$, $\therefore \cdot$ follows the commutative law